THE TYPE OF AN AFFINE FORMAL ALGEBRAIC SPACE IS INHERITED BY AN AFFINE MORPHISM OVER IT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Parts (3), (4) and (5) are proved together by the author under one heading; the card keeps that grouping and records three separate obligations from it.

Editorial note. The counted spans carry 9 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Affine formal algebraic spaces as filtered colimits of affine schemes. Same chapter, Definition definition-affine-formal-algebraic-space, invoked at proof lines 4338-4339. Supplies the presentation used in every part.
- Implications between the types of affine formal algebraic spaces, and the description of countably indexed ones as formal spectra. Same chapter, Lemma lemma-implications-between-types at proof lines 4355-4356 and 4370-4371, and Lemma lemma-countably-indexed at proof lines 4360-4362. Reduce the assumptions of parts (3) to (5) to countable indexedness and write both spaces as formal spectra.
- Morphisms between formal spectra correspond to continuous algebra homomorphisms, and a representable affine map gives a taut homomorphism. Same chapter, Lemma lemma-morphism-between-formal-spectra at proof lines 4362-4364 and Lemma lemma-representable-affine at proof lines 4364-4365. Convert the geometric map into a taut ring map so that the ascent statements apply.
- Ascent of classicity, weak adicness and the adic* property along a taut map. Same chapter, Lemmas lemma-taut-ascent-admissible, lemma-taut-ascent-weakly-adic and lemma-taut-is-adic, invoked at proof lines 4366-4368. Give parts (3), (4) and (5).
- The criterion for a Noetherian affine formal algebraic space, and local finite type of a morphism implies Noetherianness of the source for Noetherian targets. Same chapter, Lemma lemma-characterize-noetherian-affine, invoked at proof lines 4372-4373 and 4377-4378, and Morphisms, Lemma morphisms-lemma-finite-type-noetherian at proof lines 4374-4376. Give part (6).

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Source reference. Original lemma 0AKM: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2558–2558. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AN1: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2493–2493. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIK: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2453–2453. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AN0: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2355–2355. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AN5: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4239–4239. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GX8: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1436–1436. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GXI: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1741–1741. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0APU: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1557–1557. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01T6: sources/stacks-chow/tags.txt, source lines 2363–2363. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is an *affine formal algebraic space* if there exist

(1) a directed set Λ,
(2) a system $(X_\lambda, f_{\lambda\mu})$ over Λ in $(Sch/S)_{fppf}$ where

(a) each X_λ is affine,
(b) each $f_{\lambda\mu} : X_\lambda \to X_\mu$ is a thickening,

such that

$$X \cong \operatorname{colim}_{\lambda\in\Lambda} X_\lambda$$

as fppf sheaves and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of affine formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let S be a scheme. Let Y be an affine formal algebraic space. Let $f : X \to Y$ be a map of sheaves on $(Sch/S)_{fppf}$ which is representable and affine. Then*

(1) *X is an affine formal algebraic space,*
(2) *if Y is countably indexed, then X is countably indexed,*
(3) *if Y is countably indexed and classical, then X is countably indexed and classical,*
(4) *if Y is weakly adic, then X is weakly adic,*
(5) *if Y is adic*, then X is adic*, and*
(6) *if Y is Noetherian and f is (locally) of finite type, then X is Noetherian.*

Proof. Proof of (1). Write $Y = \operatorname{colim}_{\lambda\in\Lambda} Y_\lambda$ as in Definition 0.1. Since f is representable and affine, the fibre products $X_\lambda = Y_\lambda \times_Y X$ are affine. And $X = \operatorname{colim} Y_\lambda \times_Y X$. Thus X is an affine formal algebraic space.

Proof of (2). If Y is countably indexed, then in the argument above we may assume Λ is countable. Then we immediately see that X is countably indexed too.

Proof of (3), (4), and (5). In each of these cases the assumptions imply that Y is a countably indexed affine formal algebraic space (Lemma 0AIK) and hence X is too by (2). Thus we may write $X = \operatorname{Spf}(A)$ and $Y = \operatorname{Spf}(B)$ for some weakly admissible topological S-algebras A and B, see Lemma 0AN1. By Lemma 0AN0 the morphism f corresponds to a continuous S-algebra homomorphism $\varphi : B \to A$. We see from Lemma 0AN5 that φ is taut. We conclude that (3) follows from Lemma 0GX8, (4) follows from Lemma 0GXI, and (5) follows from Lemma 0APU.

Proof of (6). Combining (3) with Lemma 0AIK we see that X is adic*. Thus we can use the criterion of Lemma 0AKM. First, it tells us the affine schemes Y_λ are Noetherian. Then $X_\lambda \to Y_\lambda$ is of finite type, hence X_λ is Noetherian too (Morphisms, Lemma 01T6). Then the criterion tells us X is Noetherian and the proof is complete. □